

ML CLASSIFICATION MODEL BASED CYBER ATTACKS DETECTION

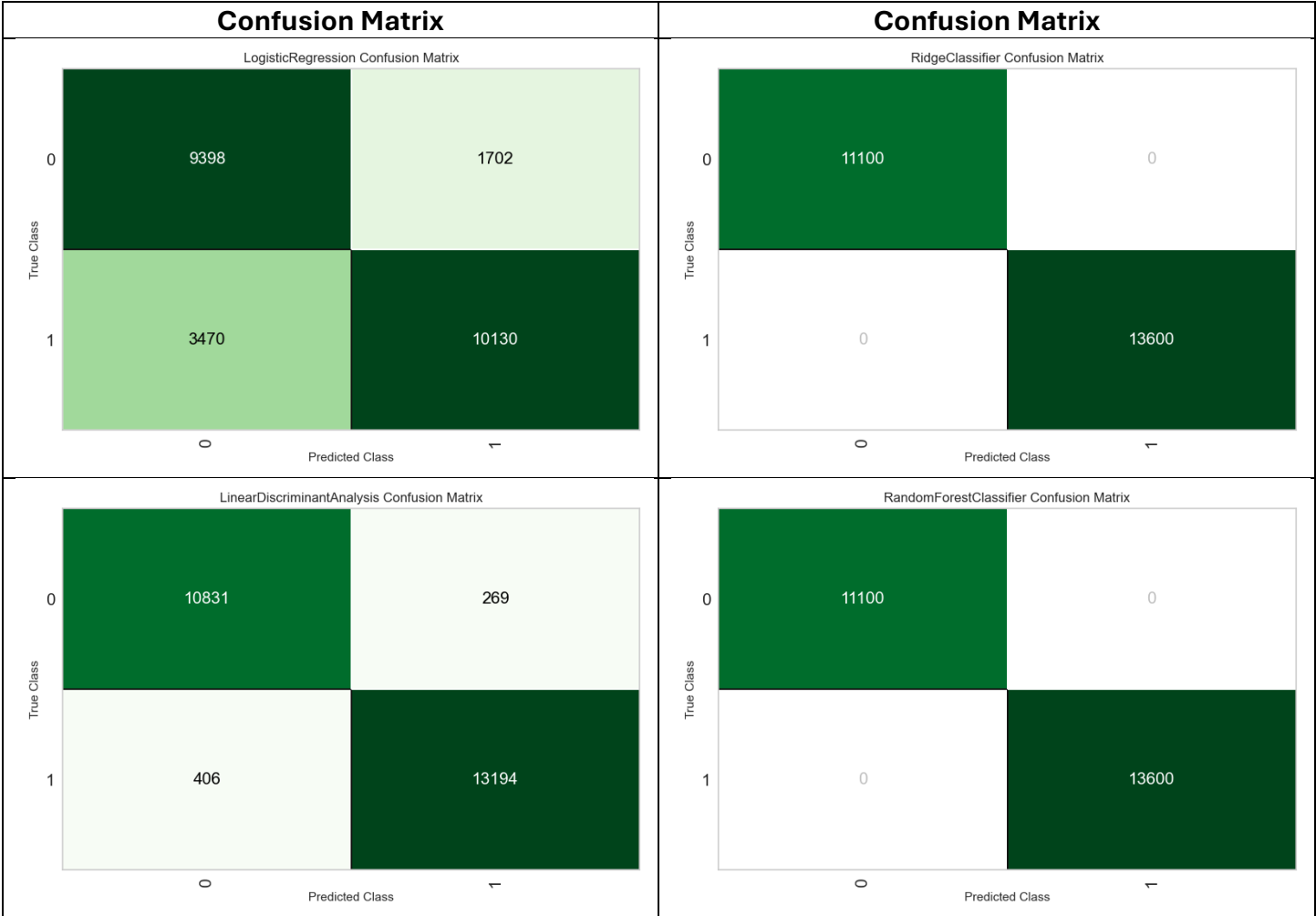
PROJECT DESCRIPTION: This project uses machine learning (ML) models to classify & detect the cyberattacks using the given indicators. The classification library from pycaret is used. The dataset contains network traffic with 42 input features which have values in normal traffic conditions and under the cyberattack conditions. The accuracy of the different models is evaluated in the report, and best or worst models are identified based on their confusion matrices, & accuracy. The data set is taken from Kaggle however the project is not based on the Kaggle competition. The data set taken was UNSW_NB15 and the code execution was done on PC. Testing and training data were cleaned to remove the row ids and cyber attack types as the cyber-attack detection & main feature identification was main purpose of the project.

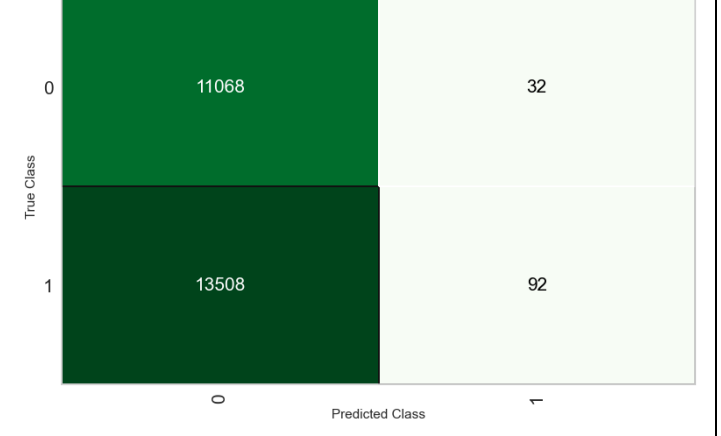
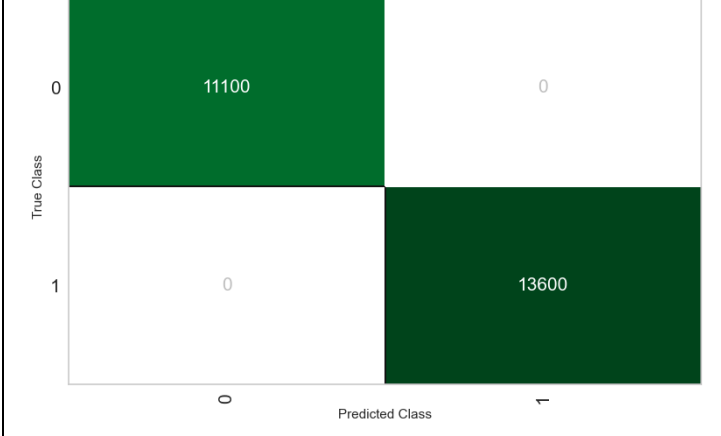
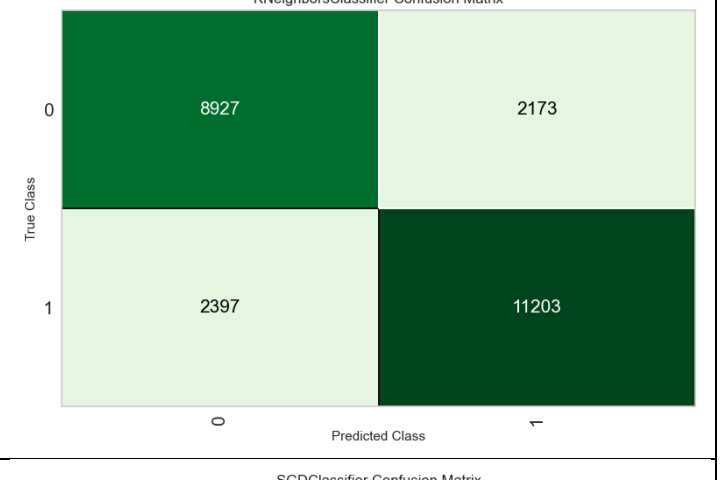
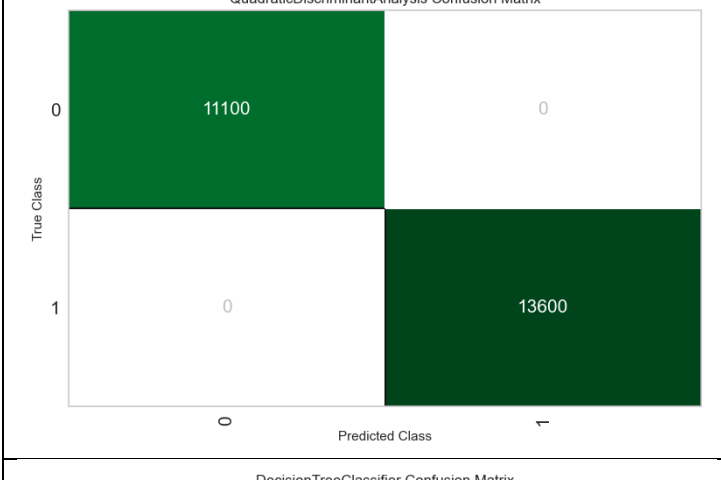
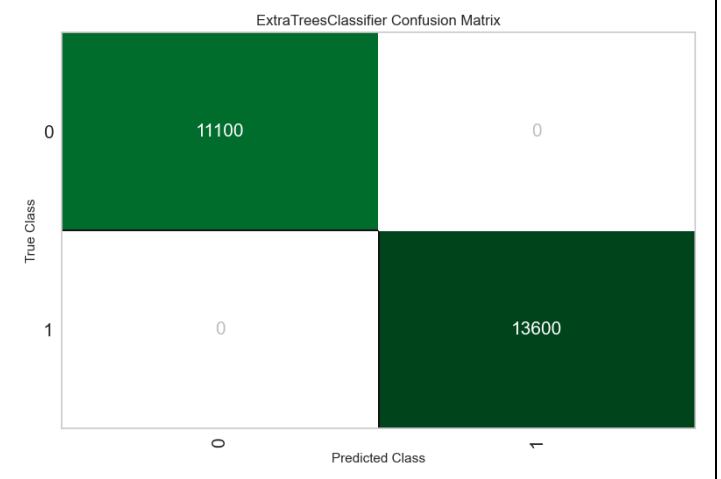
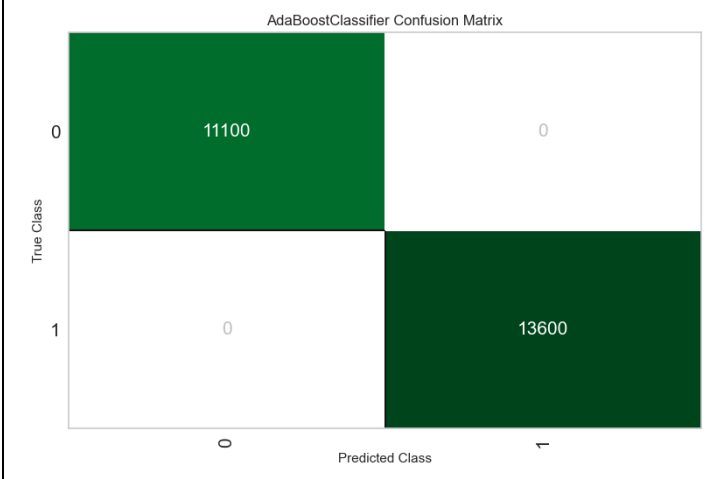
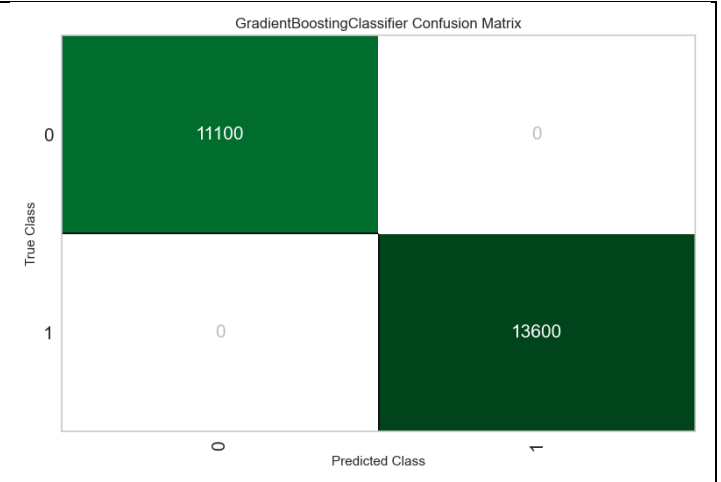
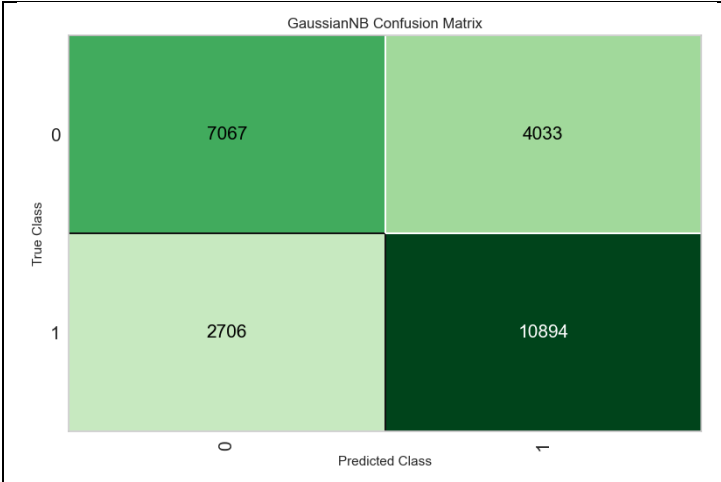
Observations

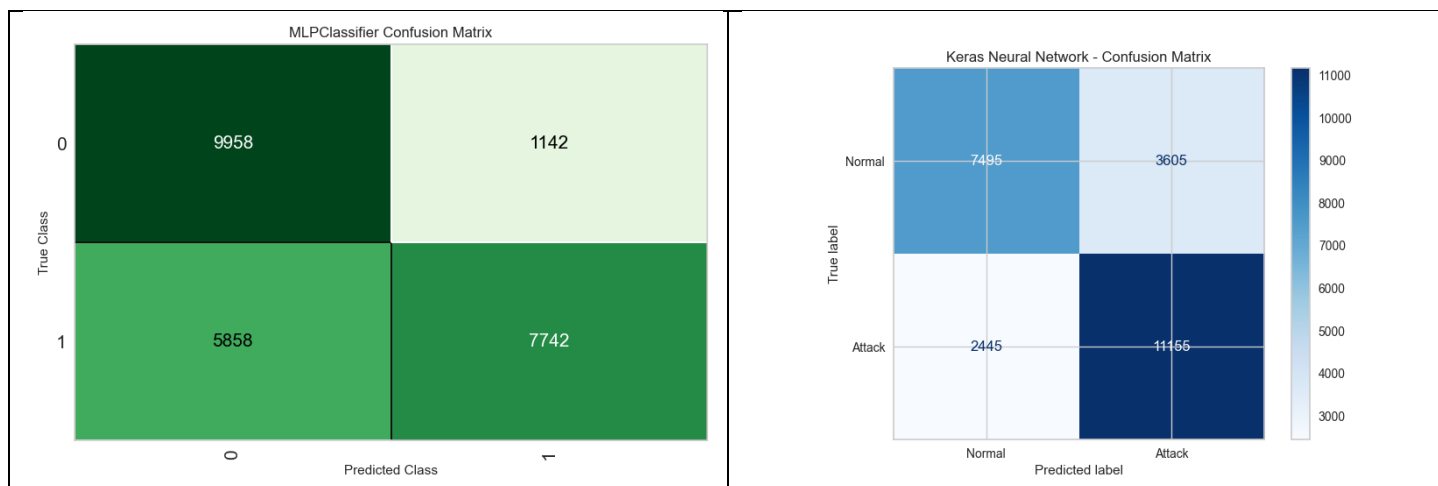
- gpc & rbfsvm models were excluded from analysis as they were very slow to process data
- lightgbm gave numerous warning messages over terminal and it has been excluded from analysis
- Each iteration when runs produces slightly change results but overall values remain similar for model accuracy, confusion matrix and features. This change might reflect for values in plots & accuracy results
- Code was stuck on simultaneous generation of plots & plots are generated by running each model one at a time for pycaret
- For feature plots, 'nb','qda','knn' & 'mlp' are not supported & gave errors during code executions
- In the pycaret installations, sgd_catboost & xgboost were not available and failed to run

Confusion Matrix

Confusion matrices are generated for the following models with their score against 0 for normal traffic and 1 for cyber attack traffic. True class value is from training set and predicted class is from testing set.







- SVM shows the worst confusion matrix results for normal & cyber attack traffics (as its underlying model is SGD, the confusion matrix header shows SGD)
- MLP & Keras NN show bad confusion matrix results for normal and cyber-attack traffics
- The Gaussian NB and K Neighbors classifiers shows significant confusion amongst the traffic

Accuracy

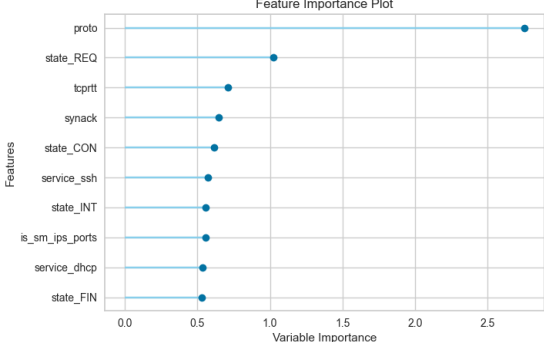
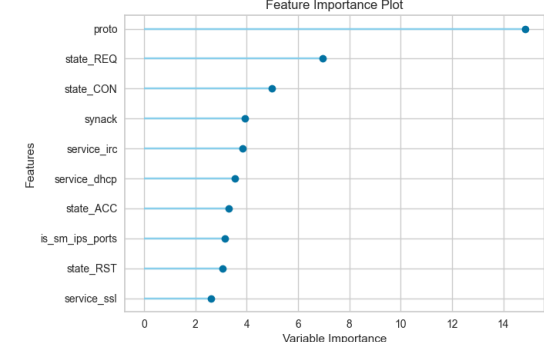
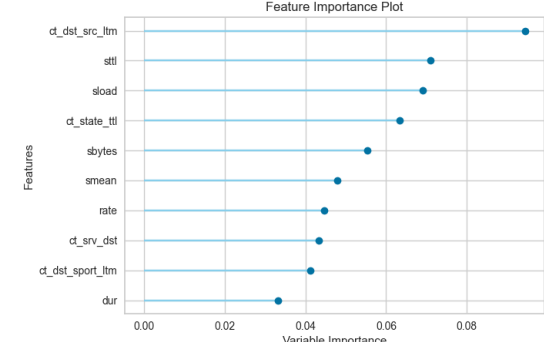
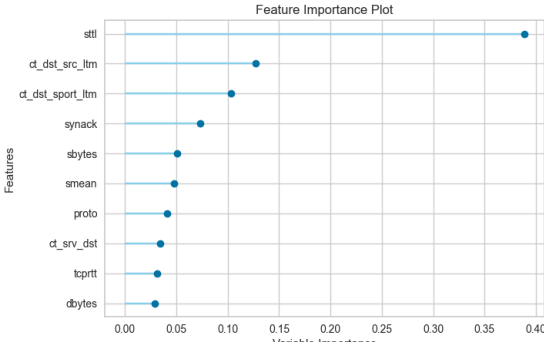
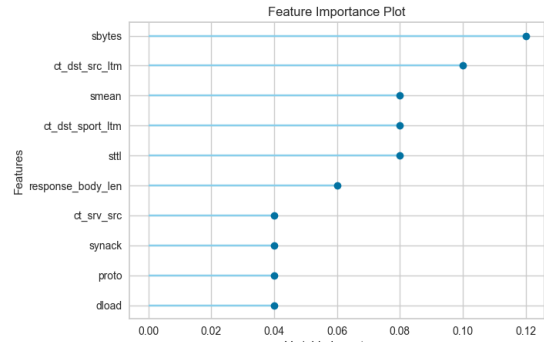
The accuracy for each model based on the normal & cyberattack traffic is presented below:

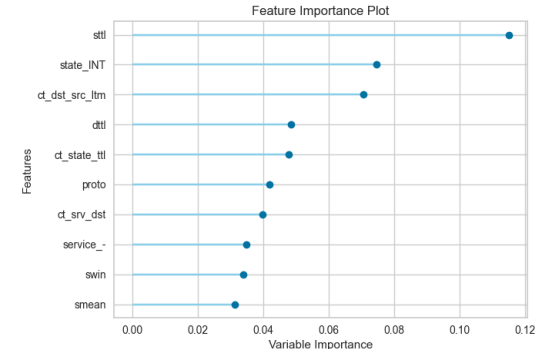
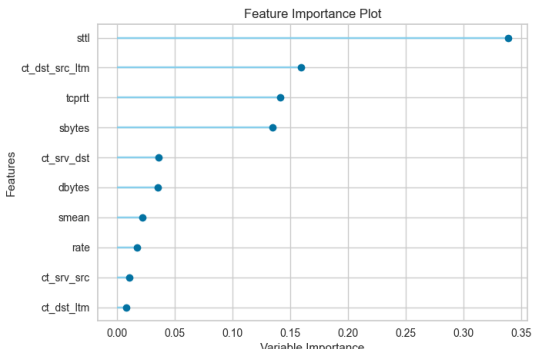
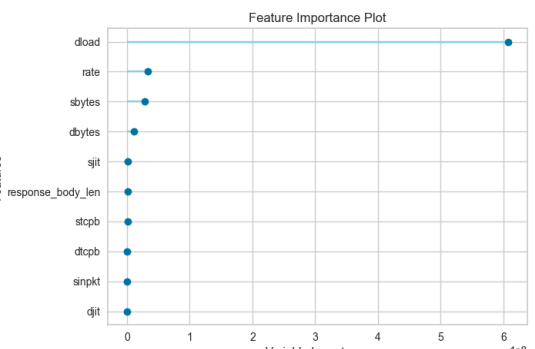
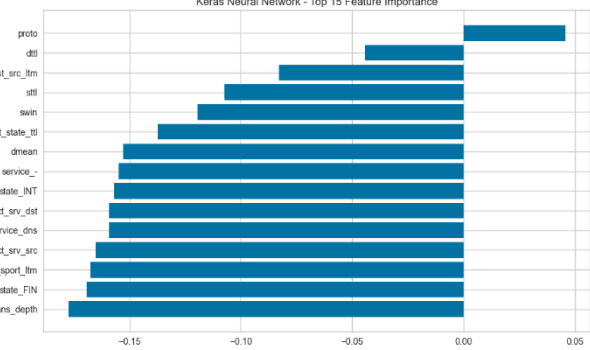
Model Name	Model Abbreviation	Accuracy (%)	Rank
Logistic regression	lr	73.86	11
Ridge	ridge	87.13	6
Linear Discriminant Analysis	lda	87.11	7
Random Forest Classifier	rf	90.29	1
Naive Bayes	nb	77.58	9
Gradient Boosting Classifier	gbc	90.12	2
Ada Boost Classifier	ada	89.62	4
Extra Trees Classifier	et	89.92	3
Quadratic Discriminant Analysis	qda	71.25	12
K Neighbors Classifier	Knn	81.16	8
Decision Tree Classifier	dt	89.58	5
SVM - Linear Kernel	svm	39.6	14
MLP Classifier	mlp	69.58	13
Keras Neural Network	Keras NN	74.19	10

Features

The features are generated for the following models with highest feature considered by each model:

Model Name	Feature	Most Important Feature
Logistic Regression	Feature Importance Plot sinpkt rate djit sbytes response_body_len dbytes dinpkt dmean sttl dload Variable Importance	Source interpacket arrival time (mSec)

Ridge	 A horizontal dot plot titled 'Feature Importance Plot' showing the importance of various features for a Ridge classifier. The y-axis lists features: proto, state_REQ, tcprrt, synack, state_CON, service_ssh, state_INT, is_sm_ips_ports, service_dhcp, and state_FIN. The x-axis is 'Variable Importance' ranging from 0.0 to 2.5. 'proto' has the highest importance at approximately 2.7, followed by 'state_REQ' at 1.0. Other features have importance values between 0.5 and 0.7. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>proto</td><td>2.7</td></tr><tr><td>state_REQ</td><td>1.0</td></tr><tr><td>tcprrt</td><td>0.7</td></tr><tr><td>synack</td><td>0.6</td></tr><tr><td>state_CON</td><td>0.6</td></tr><tr><td>service_ssh</td><td>0.55</td></tr><tr><td>state_INT</td><td>0.55</td></tr><tr><td>is_sm_ips_ports</td><td>0.55</td></tr><tr><td>service_dhcp</td><td>0.55</td></tr><tr><td>state_FIN</td><td>0.55</td></tr></table>	Features	Variable Importance	proto	2.7	state_REQ	1.0	tcprrt	0.7	synack	0.6	state_CON	0.6	service_ssh	0.55	state_INT	0.55	is_sm_ips_ports	0.55	service_dhcp	0.55	state_FIN	0.55	Transaction protocol
Features	Variable Importance																							
proto	2.7																							
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tcprrt	0.7																							
synack	0.6																							
state_CON	0.6																							
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state_INT	0.55																							
is_sm_ips_ports	0.55																							
service_dhcp	0.55																							
state_FIN	0.55																							
Linear Discriminant Analysis	 A horizontal dot plot titled 'Feature Importance Plot' showing the importance of various features for a Linear Discriminant Analysis classifier. The y-axis lists features: proto, state_REQ, state_CON, synack, service_irc, service_dhcp, state_ACC, is_sm_ips_ports, state_RST, and service_ssl. The x-axis is 'Variable Importance' ranging from 0 to 14. 'proto' has the highest importance at approximately 15, followed by 'state_REQ' at 7.5. Other features have importance values between 2 and 4. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>proto</td><td>15</td></tr><tr><td>state_REQ</td><td>7.5</td></tr><tr><td>state_CON</td><td>4.5</td></tr><tr><td>synack</td><td>4</td></tr><tr><td>service_irc</td><td>3.5</td></tr><tr><td>service_dhcp</td><td>3</td></tr><tr><td>state_ACC</td><td>2.5</td></tr><tr><td>is_sm_ips_ports</td><td>3</td></tr><tr><td>state_RST</td><td>3</td></tr><tr><td>service_ssl</td><td>2.5</td></tr></table>	Features	Variable Importance	proto	15	state_REQ	7.5	state_CON	4.5	synack	4	service_irc	3.5	service_dhcp	3	state_ACC	2.5	is_sm_ips_ports	3	state_RST	3	service_ssl	2.5	Transaction protocol
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service_irc	3.5																							
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state_ACC	2.5																							
is_sm_ips_ports	3																							
state_RST	3																							
service_ssl	2.5																							
Random Forest Classifier	 A horizontal dot plot titled 'Feature Importance Plot' showing the importance of various features for a Random Forest Classifier. The y-axis lists features: ct_dst_src_ltm, sttl, sload, ct_state_ttl, sbytes, smean, rate, ct_srv_dst, ct_dst_sport_ltm, and dur. The x-axis is 'Variable Importance' ranging from 0.00 to 0.08. 'ct_dst_src_ltm' has the highest importance at approximately 0.09, followed by 'sttl' at 0.07. Other features have importance values between 0.02 and 0.04. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>ct_dst_src_ltm</td><td>0.09</td></tr><tr><td>sttl</td><td>0.07</td></tr><tr><td>sload</td><td>0.065</td></tr><tr><td>ct_state_ttl</td><td>0.06</td></tr><tr><td>sbytes</td><td>0.05</td></tr><tr><td>smean</td><td>0.04</td></tr><tr><td>rate</td><td>0.04</td></tr><tr><td>ct_srv_dst</td><td>0.04</td></tr><tr><td>ct_dst_sport_ltm</td><td>0.04</td></tr><tr><td>dur</td><td>0.03</td></tr></table>	Features	Variable Importance	ct_dst_src_ltm	0.09	sttl	0.07	sload	0.065	ct_state_ttl	0.06	sbytes	0.05	smean	0.04	rate	0.04	ct_srv_dst	0.04	ct_dst_sport_ltm	0.04	dur	0.03	No of connections of the same source (1) and the destination (3) address in 100 connections according to the last time (26).
Features	Variable Importance																							
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ct_dst_sport_ltm	0.04																							
dur	0.03																							
Gradient Boosting Classifier	 A horizontal dot plot titled 'Feature Importance Plot' showing the importance of various features for a Gradient Boosting Classifier. The y-axis lists features: sttl, ct_dst_src_ltm, ct_dst_sport_ltm, synack, sbytes, smean, proto, ct_srv_dst, tcprrt, and dbytes. The x-axis is 'Variable Importance' ranging from 0.00 to 0.40. 'sttl' has the highest importance at approximately 0.38, followed by 'ct_dst_src_ltm' at 0.12. Other features have importance values between 0.02 and 0.08. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>sttl</td><td>0.38</td></tr><tr><td>ct_dst_src_ltm</td><td>0.12</td></tr><tr><td>ct_dst_sport_ltm</td><td>0.10</td></tr><tr><td>synack</td><td>0.08</td></tr><tr><td>sbytes</td><td>0.05</td></tr><tr><td>smean</td><td>0.05</td></tr><tr><td>proto</td><td>0.05</td></tr><tr><td>ct_srv_dst</td><td>0.04</td></tr><tr><td>tcprrt</td><td>0.04</td></tr><tr><td>dbytes</td><td>0.04</td></tr></table>	Features	Variable Importance	sttl	0.38	ct_dst_src_ltm	0.12	ct_dst_sport_ltm	0.10	synack	0.08	sbytes	0.05	smean	0.05	proto	0.05	ct_srv_dst	0.04	tcprrt	0.04	dbytes	0.04	Source to destination time to live value
Features	Variable Importance																							
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proto	0.05																							
ct_srv_dst	0.04																							
tcprrt	0.04																							
dbytes	0.04																							
Ada Boost Classifier	 A horizontal dot plot titled 'Feature Importance Plot' showing the importance of various features for an Ada Boost Classifier. The y-axis lists features: sbytes, ct_dst_src_ltm, smean, ct_dst_sport_ltm, sttl, response_body_len, ct_srv_src, synack, proto, and dload. The x-axis is 'Variable Importance' ranging from 0.00 to 0.12. 'sbytes' has the highest importance at approximately 0.11, followed by 'ct_dst_src_ltm' at 0.10. Other features have importance values between 0.04 and 0.08. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>sbytes</td><td>0.11</td></tr><tr><td>ct_dst_src_ltm</td><td>0.10</td></tr><tr><td>smean</td><td>0.08</td></tr><tr><td>ct_dst_sport_ltm</td><td>0.08</td></tr><tr><td>sttl</td><td>0.08</td></tr><tr><td>response_body_len</td><td>0.06</td></tr><tr><td>ct_srv_src</td><td>0.04</td></tr><tr><td>synack</td><td>0.04</td></tr><tr><td>proto</td><td>0.04</td></tr><tr><td>dload</td><td>0.04</td></tr></table>	Features	Variable Importance	sbytes	0.11	ct_dst_src_ltm	0.10	smean	0.08	ct_dst_sport_ltm	0.08	sttl	0.08	response_body_len	0.06	ct_srv_src	0.04	synack	0.04	proto	0.04	dload	0.04	Source to destination transaction bytes
Features	Variable Importance																							
sbytes	0.11																							
ct_dst_src_ltm	0.10																							
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ct_srv_src	0.04																							
synack	0.04																							
proto	0.04																							
dload	0.04																							

Extra Trees Classifier	Feature Importance Plot  This plot shows the variable importance for the Extra Trees Classifier. The x-axis represents 'Variable Importance' from 0.00 to 0.12. The y-axis lists features. 'sttl' is the most important feature, followed by 'state_INT' and 'ct_dst_src_ttm'. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>sttl</td><td>0.115</td></tr><tr><td>state_INT</td><td>0.075</td></tr><tr><td>ct_dst_src_ttm</td><td>0.070</td></tr><tr><td>dttl</td><td>0.050</td></tr><tr><td>ct_state_ttl</td><td>0.045</td></tr><tr><td>proto</td><td>0.040</td></tr><tr><td>ct_srv_dst</td><td>0.035</td></tr><tr><td>service_</td><td>0.030</td></tr><tr><td>swin</td><td>0.025</td></tr><tr><td>smean</td><td>0.020</td></tr></table>	Features	Variable Importance	sttl	0.115	state_INT	0.075	ct_dst_src_ttm	0.070	dttl	0.050	ct_state_ttl	0.045	proto	0.040	ct_srv_dst	0.035	service_	0.030	swin	0.025	smean	0.020	Source to destination time to live value										
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Decision Tree Classifier	Feature Importance Plot  This plot shows the variable importance for the Decision Tree Classifier. The x-axis represents 'Variable Importance' from 0.00 to 0.35. The y-axis lists features. 'sttl' is the most important feature, followed by 'ct_dst_src_ttm' and 'tcprtt'. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>sttl</td><td>0.340</td></tr><tr><td>ct_dst_src_ttm</td><td>0.160</td></tr><tr><td>tcprtt</td><td>0.140</td></tr><tr><td>sbytes</td><td>0.130</td></tr><tr><td>ct_srv_dst</td><td>0.030</td></tr><tr><td>dbytes</td><td>0.025</td></tr><tr><td>smean</td><td>0.020</td></tr><tr><td>rate</td><td>0.015</td></tr><tr><td>ct_srv_src</td><td>0.010</td></tr><tr><td>ct_dst_ttm</td><td>0.005</td></tr></table>	Features	Variable Importance	sttl	0.340	ct_dst_src_ttm	0.160	tcprtt	0.140	sbytes	0.130	ct_srv_dst	0.030	dbytes	0.025	smean	0.020	rate	0.015	ct_srv_src	0.010	ct_dst_ttm	0.005	Source to destination time to live value										
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ct_dst_ttm	0.005																																	
SVM	Feature Importance Plot  This plot shows the variable importance for the SVM classifier. The x-axis represents 'Variable Importance' from 0 to 6e8. The y-axis lists features. 'dload' is the most important feature, followed by 'rate' and 'sbytes'. <table><tr><th>Features</th><th>Variable Importance</th></tr><tr><td>dload</td><td>6.0e8</td></tr><tr><td>rate</td><td>0.5e8</td></tr><tr><td>sbytes</td><td>0.4e8</td></tr><tr><td>dbytes</td><td>0.3e8</td></tr><tr><td>sjit</td><td>0.2e8</td></tr><tr><td>response_body_len</td><td>0.1e8</td></tr><tr><td>stcpb</td><td>0.1e8</td></tr><tr><td>drcpb</td><td>0.1e8</td></tr><tr><td>sinpkt</td><td>0.1e8</td></tr><tr><td>djit</td><td>0.1e8</td></tr></table>	Features	Variable Importance	dload	6.0e8	rate	0.5e8	sbytes	0.4e8	dbytes	0.3e8	sjit	0.2e8	response_body_len	0.1e8	stcpb	0.1e8	drcpb	0.1e8	sinpkt	0.1e8	djit	0.1e8	Destination bits per second										
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Keras NN	Keras Neural Network - Top 15 Feature Importance  This plot shows the top 15 feature importance for the Keras Neural Network. The x-axis represents 'Drop in Accuracy when Feature is Shuffled' from -0.15 to 0.05. The y-axis lists features. 'proto' is the most important feature, followed by 'dttl' and 'ct_dst_src_ttm'. <table><tr><th>Features</th><th>Drop in Accuracy when Feature is Shuffled</th></tr><tr><td>proto</td><td>0.045</td></tr><tr><td>dttl</td><td>0.040</td></tr><tr><td>ct_dst_src_ttm</td><td>0.035</td></tr><tr><td>sttl</td><td>0.030</td></tr><tr><td>swin</td><td>0.025</td></tr><tr><td>ct_state_ttl</td><td>0.020</td></tr><tr><td>dmean</td><td>0.015</td></tr><tr><td>service_</td><td>0.010</td></tr><tr><td>state_INT</td><td>0.005</td></tr><tr><td>ct_srv_dst</td><td>0.000</td></tr><tr><td>service_dns</td><td>-0.005</td></tr><tr><td>ct_srv_src</td><td>-0.010</td></tr><tr><td>ct_dst_sport_ttm</td><td>-0.015</td></tr><tr><td>state_FIN</td><td>-0.020</td></tr><tr><td>trans_depth</td><td>-0.025</td></tr></table>	Features	Drop in Accuracy when Feature is Shuffled	proto	0.045	dttl	0.040	ct_dst_src_ttm	0.035	sttl	0.030	swin	0.025	ct_state_ttl	0.020	dmean	0.015	service_	0.010	state_INT	0.005	ct_srv_dst	0.000	service_dns	-0.005	ct_srv_src	-0.010	ct_dst_sport_ttm	-0.015	state_FIN	-0.020	trans_depth	-0.025	Represents the pipelined depth into the connection of http request/response transaction
Features	Drop in Accuracy when Feature is Shuffled																																	
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ct_srv_src	-0.010																																	
ct_dst_sport_ttm	-0.015																																	
state_FIN	-0.020																																	
trans_depth	-0.025																																	

The following features were considered most important by the models:

Model Name	Feature	Accuracy (%)	Confusion Matrix	Analysis
Logistic Regression	Source interpacket arrival time (mSec)	73.86	Confusion observed	Slightly Lower accuracy
Ridge	Transaction protocol	87.13	No confusion	Transaction protocol is identified as an indication of cyber-attack with high accuracy models
Linear Discriminant Analysis	Transaction protocol	87.13	Slight confusion	
Random Forest Classifier	No of connections of the same source (1) and the destination (3) address in 100 connections according to the last time (26).	90.29	No confusion	Same source destination IPs indicate denial of service attacks
Gradient Boosting Classifier	Source to destination time to live value	90.12	No Confusion	Time to live is an indication for the pack to stay alive before it is discarded
Ada Boost Classifier	Source to destination transaction bytes	89.62	No confusion	
Extra Trees Classifier	Source to destination time to live value	89.92	No confusion	
Decision Tree Classifier	Source to destination time to live value	89.58	No confusion	
SVM	Destination bits per second	39.6	High confusion	Low accuracy
Keras NN	Represents the pipelined depth into the connection of http request/response transaction	74.19	Confusion observed	Slightly lower accuracy

Conclusion:

A clear pattern has emerged after the analysis of model features, confusion matrix and accuracy. The top three features identified for the dataset with the highest accuracy models are **transaction protocols, same source and destination IPs, and packet time to live**. These three features can be used to detect cyber-attacks in the network using the highest accuracy models. **Random forest, gradient boosting and extra trees** are the top three models observed based on accuracy.